Customer Churn & Retention Analytics

1. Project Objective

The objective of this project is to understand why customers leave a telecommunications service. The analysis examines customer demographics, services, contracts, billing, and churn patterns. The findings can help the company identify customers who may need retention support.

2. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI Desktop, Power Query, and DAX.

3. Dataset Description

The dataset contains customer and service information, including Customer ID, Gender, Senior Citizen, Partner, Dependents, Tenure Months, Internet Service, Contract, Payment Method, Monthly Charges, Total Charges, and Churn.

4. Data Cleaning

The dataset was loaded into Power BI and checked in Power Query. Customer ID was kept as Text. Tenure Months and charge fields were checked for suitable numeric data types. The data was then applied to the Power BI report.

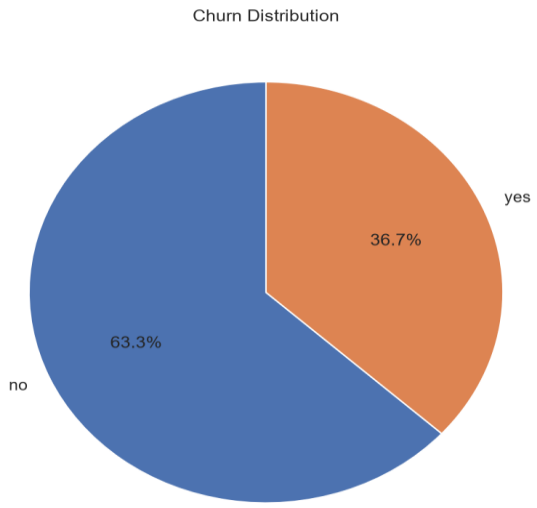
5. Data Analysis

DAX measures were used to calculate total customers, churned customers, retained customers, churn rate, average monthly charges, and average tenure. The measures use functions such as DISTINCTCOUNT, COUNTROWS, CALCULATE, DIVIDE, and AVERAGE.

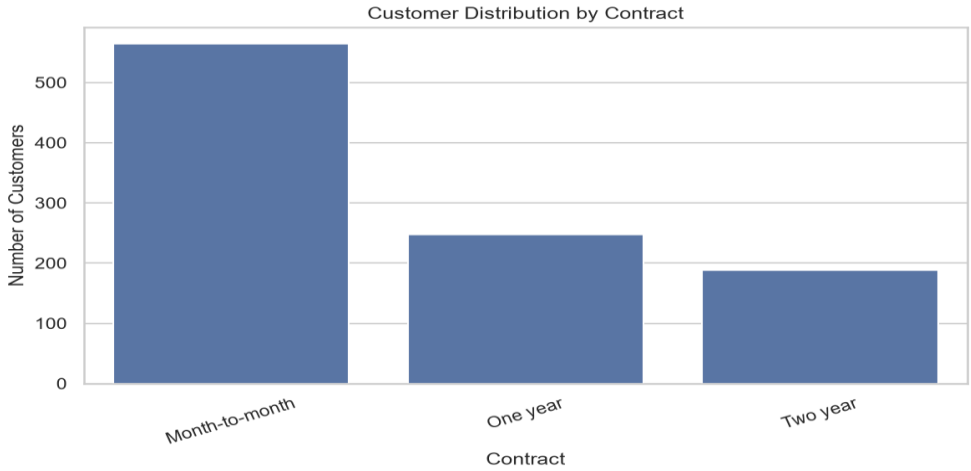
6. Python Visualizations

The following visualizations can be created with Matplotlib and Seaborn:

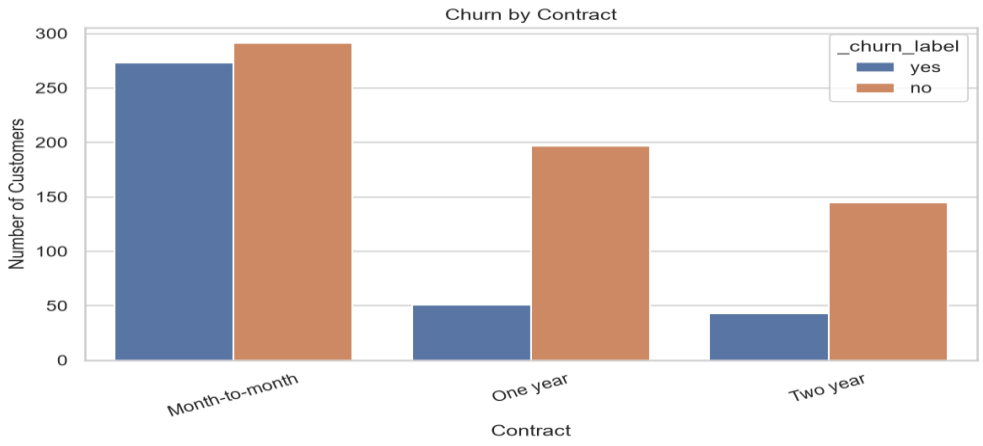
1. **Churn distribution**



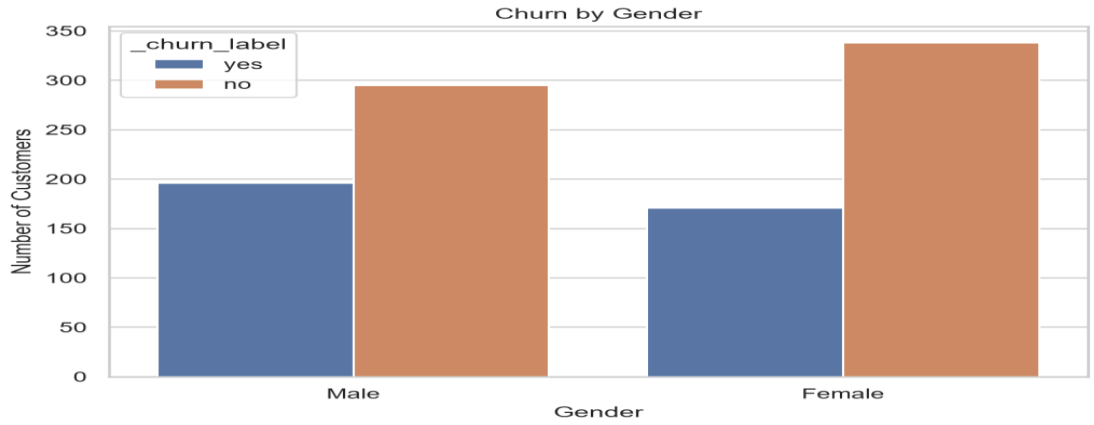
2. Customer count by Contract



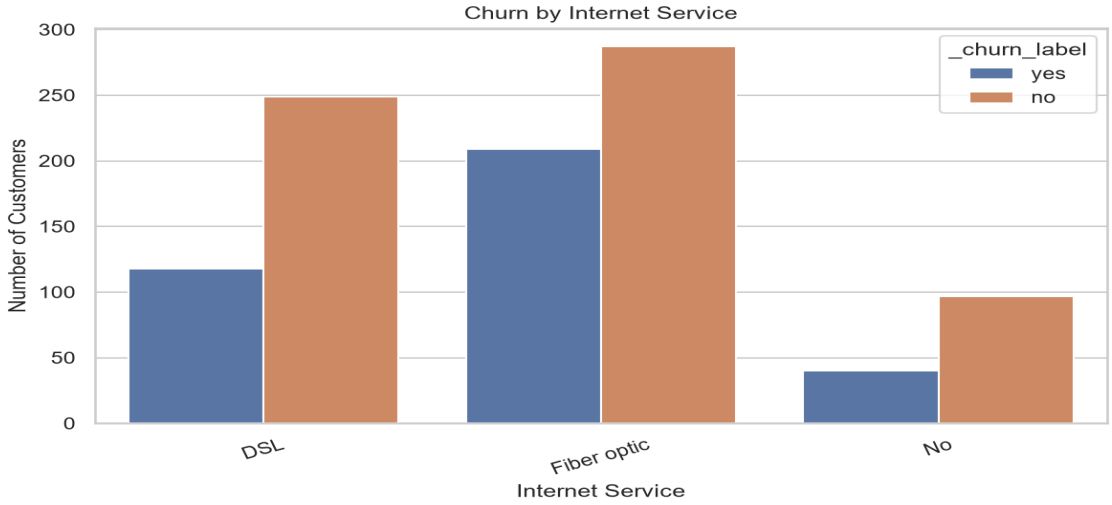
3. Churn by Contract



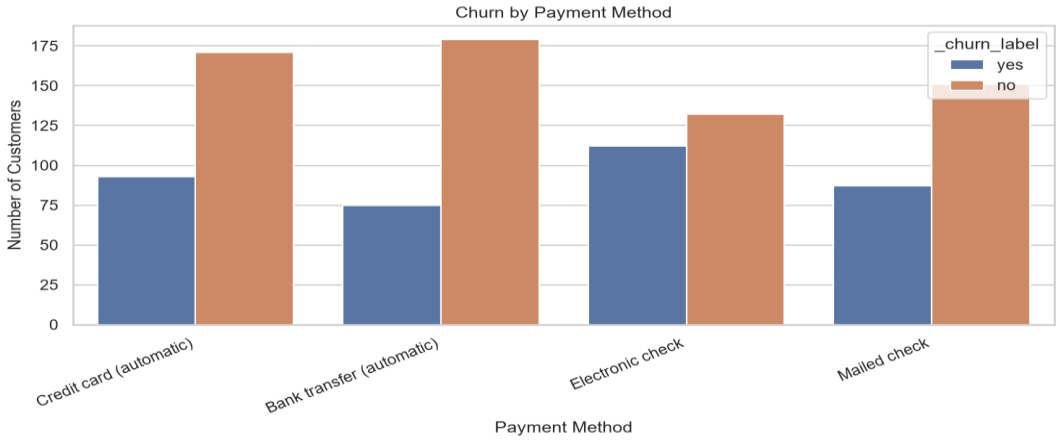
4. Churn by Gender



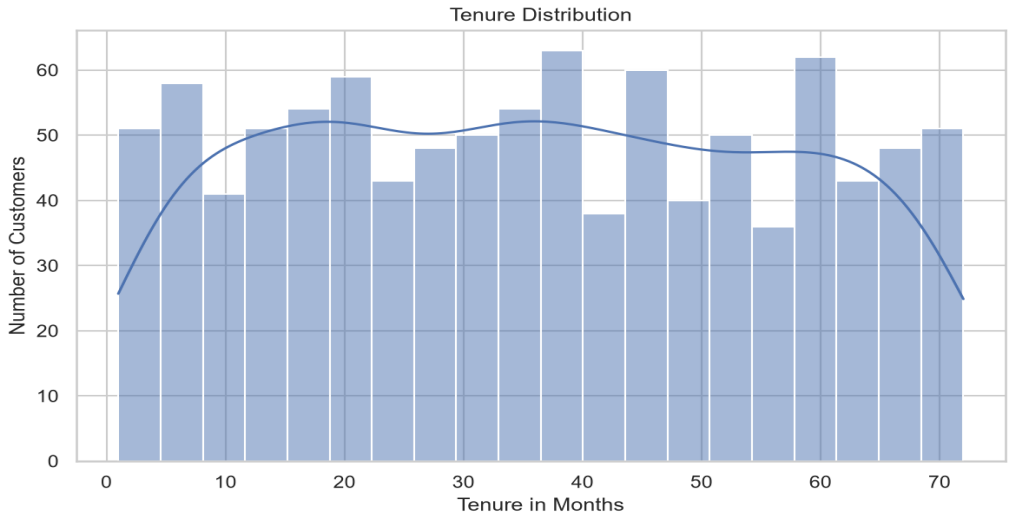
5. Churn by Internet Service



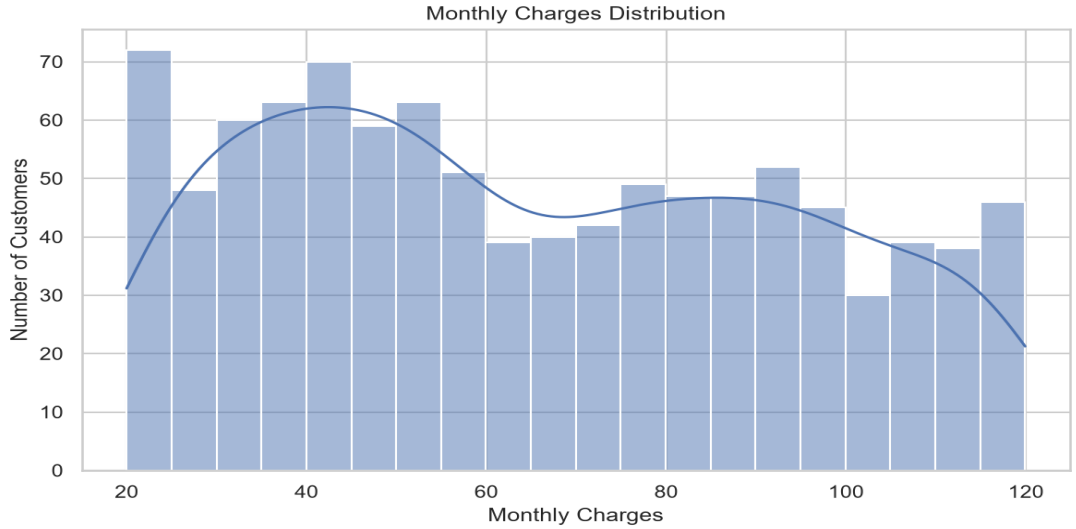
6. Churn by Payment Method



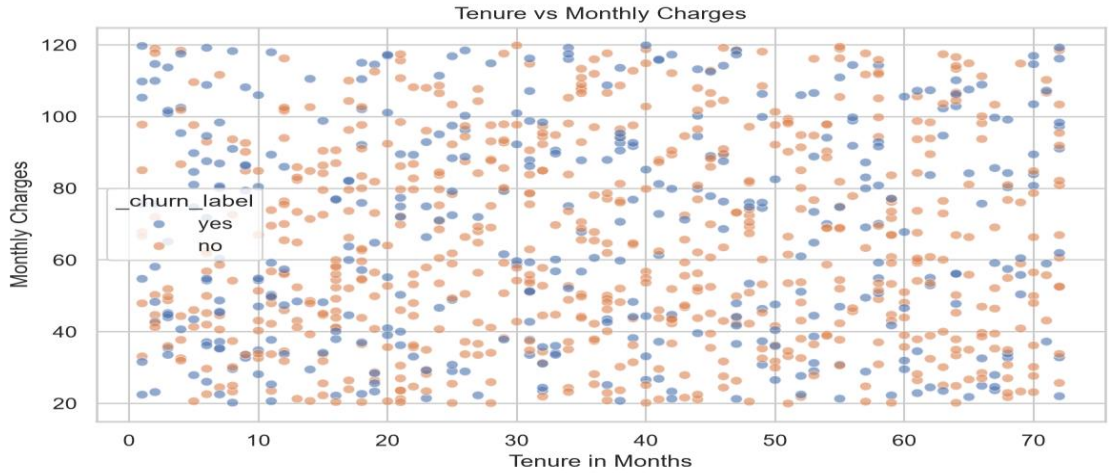
7. Tenure distribution



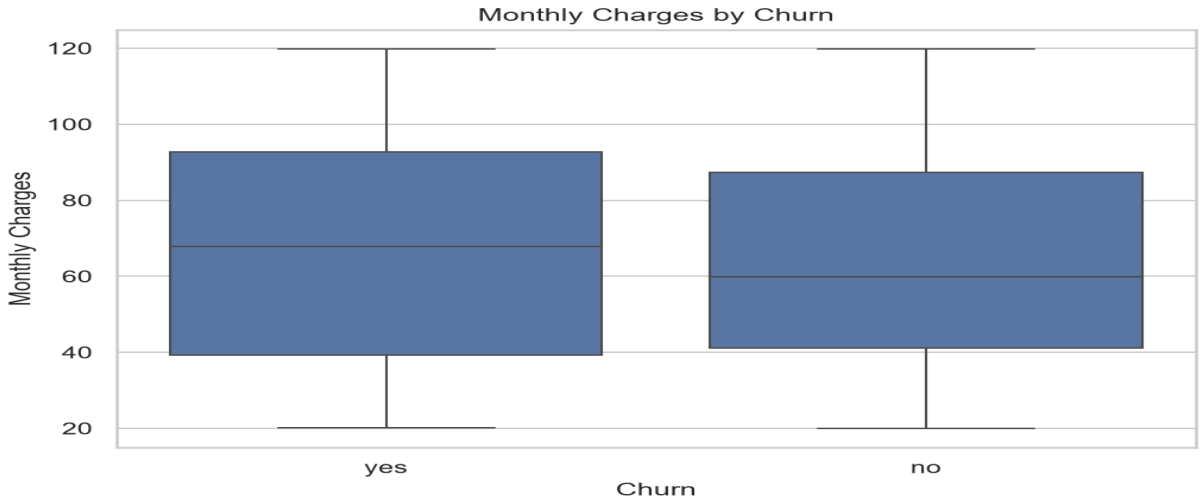
8. Monthly Charges distribution



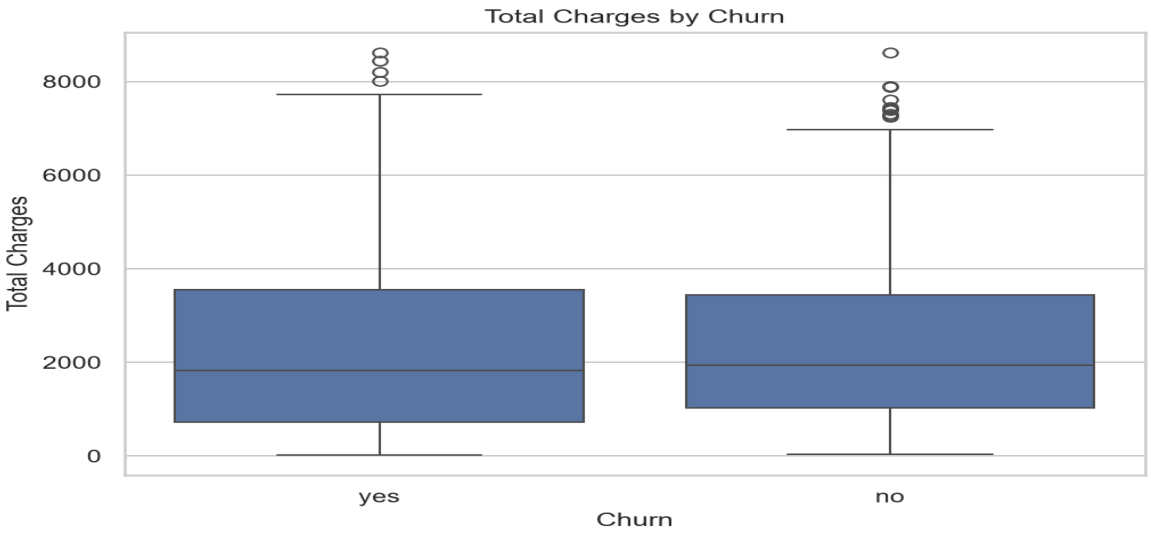
9. Tenure vs. Monthly Charges



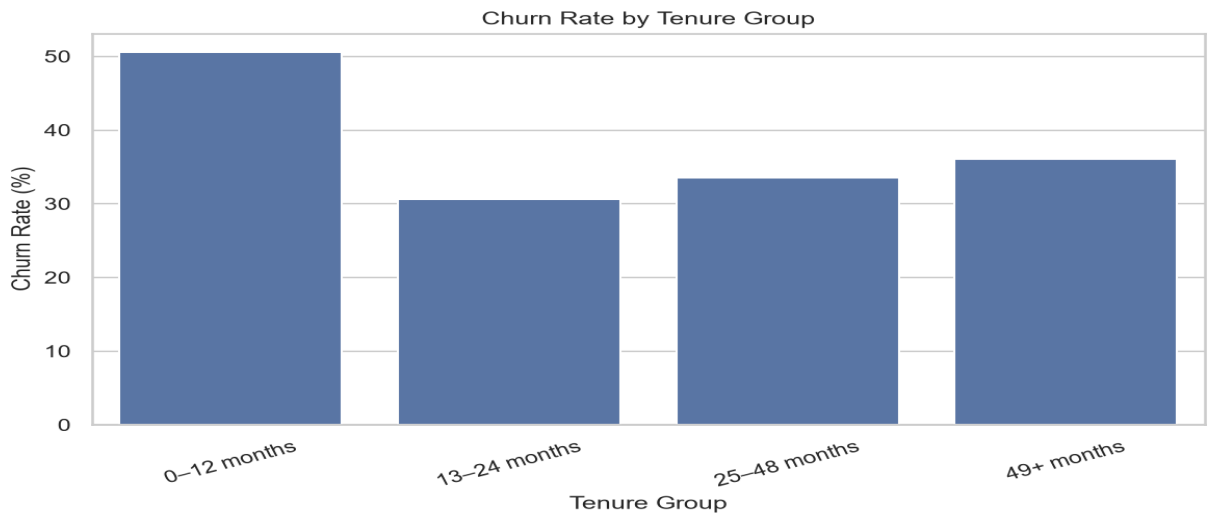
10. Monthly Charges by Churn



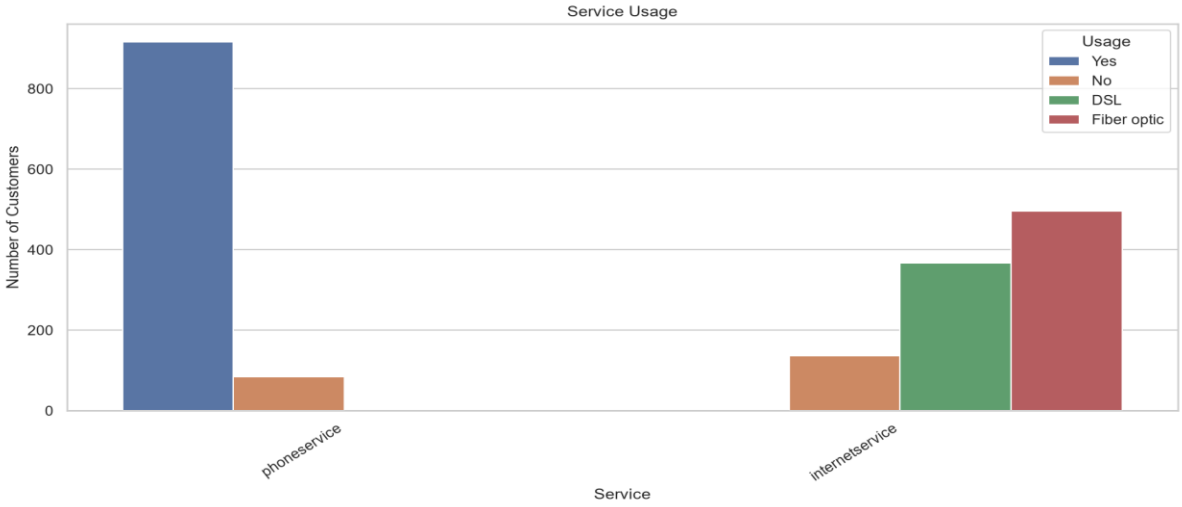
11. Total Charges by Churn



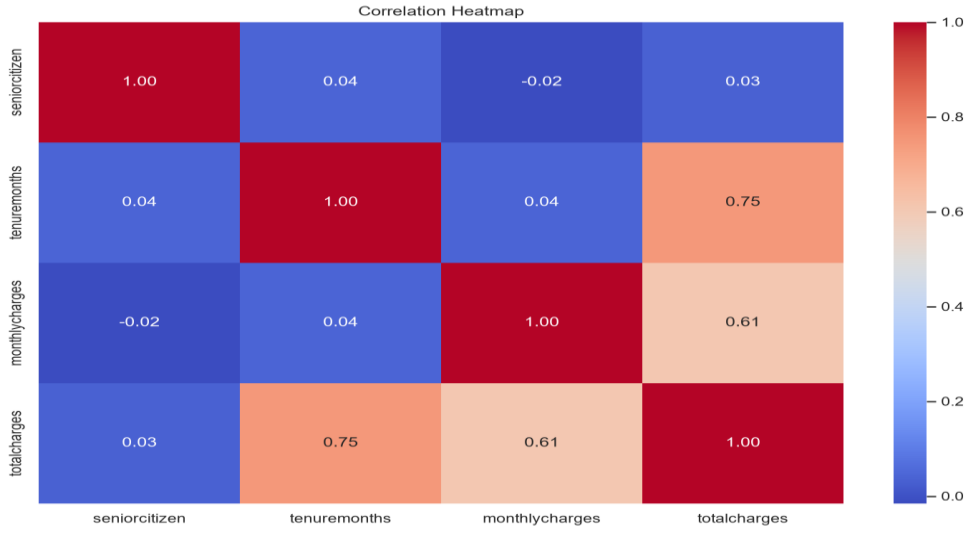
12. Churn Rate by Tenure Group



13. Service usage



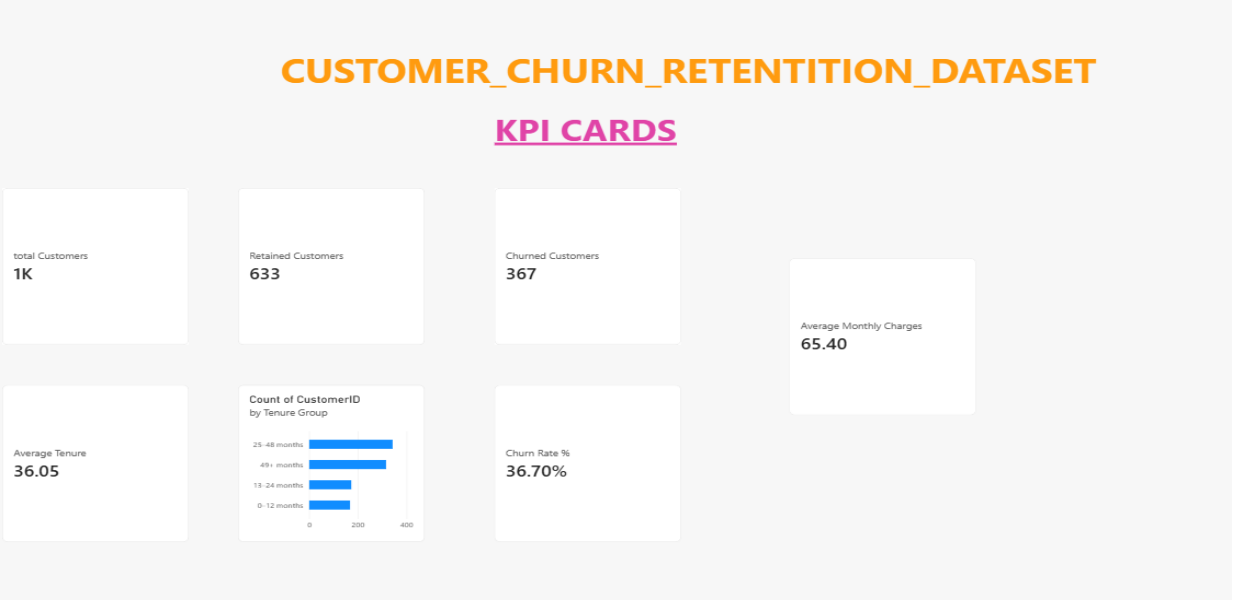
14. Correlation heatmap



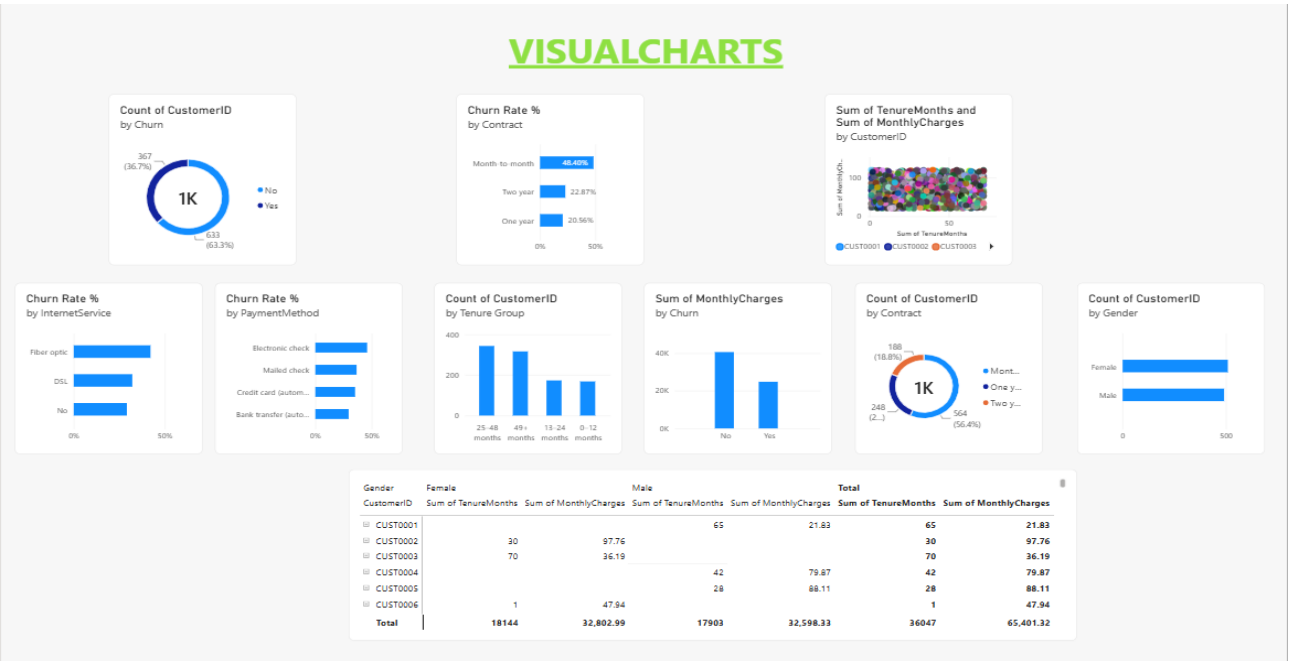
6. Power BI Dashboard

The interactive dashboard contains KPI Cards, charts, customer details, and slicers for Gender, Contract, Internet Service, Payment Method, Senior Citizen, and Churn. The slicers allow users to filter the report.

KPI CARDS

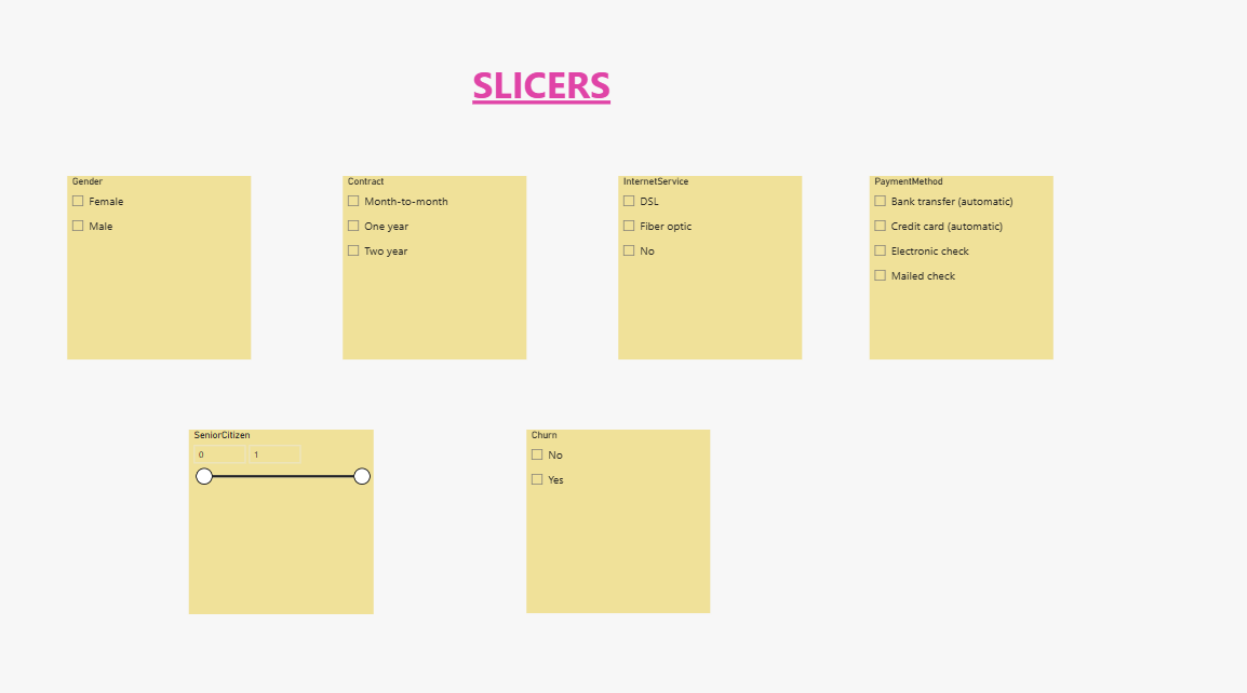


VISUAL CHARTS



Gender	Female		Male		Total	
CustomerID	Sum of TenureMonths	Sum of MonthlyCharges	Sum of TenureMonths	Sum of MonthlyCharges	Sum of TenureMonths	Sum of MonthlyCharges
CUST0001			65	21.83	65	21.83
CUST0002	30	97.76			30	97.76
CUST0003	70	36.19			70	36.19
CUST0004			42	79.87	42	79.87
CUST0005			28	88.11	28	88.11
CUST0006	1	47.94			1	47.94
Total	18144	32,802.99	17903	32,598.33	36047	65,401.32

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7. Key Insights

- The dashboard shows **1,000 total customers**.
- **367 customers churned** and **633 customers were retained**.
- The overall churn rate is **36.70%**.
- The Contract chart shows the highest churn rate for **Month-to-month** customers.
- The Payment Method chart shows the highest churn rate for **electronic check** customers.
- The Internet Service chart shows the highest churn rate for **Fiber optic** customers.
- The overall average monthly charge is **65.40**, and the average tenure is **36.05 months**.
- Compare churn rates by tenure group to confirm how tenure relates to churn. Use **Average** (not Sum) for Monthly Charges when comparing charges by Churn.

8. Conclusion

The dashboard indicates that month-to-month customers, electronic-check users, and fiber-optic customers are important groups to review for retention. The company can investigate these groups further and offer suitable support or plans to improve customer retention.